Modeling Flexible Business Processes with Optional and Alternative Activities: A Unified BPMN Approach to Loan Application Handling

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***Abstract*—Business process models usually prescribe a single sequence of activities, yet organizations execute the same process differently across branches, customer segments and risk categories. Keeping one diagram for every variant leads to duplicated, hard-to-maintain model collections. This report studies how *optional* and *alternative* activities can be used to fold several rigid process variants into one BPMN model through design-time flexibility by selection. Using a loan application process as an illustrative case, four realistic variants (standard branch, digital fast-track, legacy paper-file and high-risk rejected loans) are analysed and merged into a single model built from labelled exclusive (XOR) gateways and one bypass for the optional archiving step. Tracing shows that all four variants remain executable. Structurally, the number of diagrams falls from four to one and the activity count from 22 to 9, at the price of eight additional XOR gateways and their guard conditions. The evaluation is structural and illustrative; empirical usability studies and controlled-flexibility annotations are left for future work.**

***Index Terms*—business process management, BPMN, process flexibility, flexibility by selection, optional activities, alternative activities, process variants**

I. Introduction

Process models are the common language through which organizations document, analyse and automate their operations. Conventional activity-oriented notations, however, fix the order of tasks: every case is expected to travel along the same path. Real operations rarely behave this way. Regional offices apply local rules, products differ in their risk profile, and some tasks are required for one customer type but irrelevant for another.

Two unsatisfactory responses are common. The first is to draw a separate model for every variant, which multiplies the artefacts that must be kept consistent. The second is to keep a single idealised model and leave deviations to the discretion of the participants, which removes guidance and weakens control. Flexibility that is built into the model in advance offers a middle course: the variation is anticipated, drawn explicitly, and therefore remains verifiable.

This report concentrates on that middle course. It examines how the two simplest flexibility constructs, optional activities and alternative activities, can be expressed in BPMN 2.0 [7] and how they allow a loan application process with four variants to be captured in one model. The contributions of the work are:

- a classification of the variation present in four loan-handling variants into shared, alternative and optional activities;
- a unified BPMN model, together with its rigid baseline, that covers all four variants using labelled XOR gateways and a bypass flow;
- a structural comparison of the separate and unified representations, including the costs introduced by the unification.

The rest of the report is organised as follows. Section II summarises the background literature, Section III states the problem and research question, Section IV presents the methodology, Section V reports the results, Section VI discusses limitations, and Section VII concludes with directions for future work.

II. Background and Related Work

A. Flexibility by Selection and by Adaptation

Nurcan [1] surveys the flexibility requirements placed on business processes and their modeling artefacts and separates two families of flexibility. *Flexibility by selection* (a priori) offers a set of pre-modeled alternatives from which the executing case picks. *Flexibility by adaptation* (a posteriori) modifies the model or a running instance while it is being executed. The former is available at build time and keeps every case on a path that was verified in advance; the latter handles unforeseen events but raises the problem of migrating running instances. The present work belongs entirely to the first family. A related taxonomy of flexibility types is given by Regev et al. [4].

B. Configurable Process Models

Configurable models describe a family of related processes in a single artefact. Rosemann and van der Aalst [6] propose a configurable reference modeling language, and Gottschalk et al. [5] extend workflow models with configuration points. Buijs et al. [2] discuss how configurable models are obtained from collections of event logs, where a branch of the model can be *hidden*, which corresponds to an optional activity, or

Step 1 Define variants → **Step 2** Classify activities → **Step 3** Build one model → **Step 4** Validate & compare

Fig. 1. Overview of the proposed methodology.

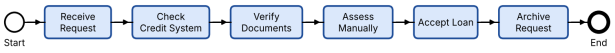


Fig. 2. Rigid BPMN model of variant V1 (standard branch loan). Variants V2–V4 are drawn in the same way with their own task sequences.

blocked, which corresponds to choosing one alternative over another. These two operations motivate the optional/alternative distinction used here.

C. Guidance in Flexible Models

Flexibility without guidance can mislead participants. Martinho et al. [3] introduce CF4BPMN, a BPMN extension in which each permitted change is classified as allowed, discouraged or denied, so that users receive explicit direction on admissible choices. This idea is included in the plan for the next phase of the project (Section VII).

III. Problem Statement and Research Question

Three issues recur in the literature reviewed above. **(i)** Maintaining one model per variant causes redundancy [2]. **(ii)** Participants need explicit rules about which choices are valid [3]. **(iii)** Activity-oriented notations tend to prescribe a strict order of steps [1].

Based on these issues, the research question addressed in this work is:

How can optional and alternative activities be modeled so that a single unified process remains simple, covers the real process variants, and still guides its users?

IV. Methodology

The method is a four-step procedure that turns a set of rigid variants into a single flexible BPMN model. Fig. 1 summarises the sequence and the sections below describe each step for the loan case.

A. Step 1: Defining the Process Variants

The loan application process is divided into six stages: intake, credit check, document verification, assessment, decision, and archiving. Four variants, listed in Table I, arise from the way branches and loan categories carry out these stages.

Every variant is first drawn as a rigid model. The baseline for V1, a strictly sequential chain of six tasks, is shown in Fig. 2.

B. Step 2: Classifying Activities

The activities of the four variants are compared stage by stage and placed into three classes.

- **Shared (mandatory) activities** occur in every variant: *Receive Request* and *Verify Documents*.
- **Alternative activities** fulfil the same purpose in different ways, so exactly one of them is selected per case. Three pairs were found: *Check Credit System* / *Check Paper Archive*, *Assess Manually* / *Auto-Approve Check*, and *Accept Loan* / *Reject Loan*.
- **Optional activity**: *Archive Request* appears in V1 and V3 and is omitted from V2 and V4, so its omission does not invalidate the case.

TABLE I
Loan Application Variants (activities per stage)

ID	Credit	Assessment	Decision	Archive	#
V1 Standard branch	Credit system	Manual	Accept	Archive	6
V2 Digital fast-track	Credit system	Auto-approve	Accept	Skipped	5
V3 Legacy paper-file	Paper archive	Manual	Accept	Archive	6
V4 High-risk rejected	Credit system	Manual	Reject	Skipped	5

All variants begin with *Receive Request* and include *Verify Documents*; the # column counts activities per variant.

This gives $2 + 6 + 1 = 9$ unique activities across the family, compared with 22 activity occurrences in the separate models.

C. Step 3: Building the Unified Model

Two BPMN patterns are used, both based on the exclusive (XOR) gateway.

1) *Alternative pattern:* a diverging XOR gateway routes each case to exactly one of two tasks, and a converging XOR gateway merges the branches again.
2) *Optional pattern:* a diverging XOR gateway offers either the task or a direct bypass flow, and a converging gateway merges them.

Each diverging gateway has one *default* flow and, on the other branch, a guard condition. The model is written in BPMN 2.0 XML, and the conditions are FEEL expressions on a process variable `variant` whose value (V1–V4) identifies the intended path. Table II lists the four variation points.

D. Step 4: Validation and Comparison

Validation has two parts. First, each of the four variants is traced through the unified model by setting `variant` and following the gateway conditions; the resulting activity sequence must equal the sequence of the corresponding rigid model. Second, the separate and unified representations are compared using simple counts of BPMN elements: diagrams, activity nodes, start/end events, XOR gateways, and sequence flows. The three evaluation criteria are *path coverage* (all

TABLE II
VARIATION POINTS AND GATEWAY CONFIGURATION

Gateway	Type	Default branch	Guarded branch (condition)
Credit Source	Alternative	Check Credit System	Check Paper Archive (V3)
Assessment Mode	Alternative	Assess Manually	Auto-Approve Check (V2)
Loan Decision	Alternative	Accept Loan	Reject Loan (V4)
Archive?	Optional	Skip (bypass)	Archive Request (V1 or V3)

TABLE III
TRACE OF VARIANTS THROUGH THE UNIFIED MODEL

Variant	Credit	Assess	Decision	Archive	Tasks	Match
V1	System	Manual	Accept	Yes	6	✓
V2	System	Auto	Accept	Bypass	5	✓
V3	Paper	Manual	Accept	Yes	6	✓
V4	System	Manual	Reject	Bypass	5	✓

variants stay executable), *structural size*, and the *modeling trade-off* between less duplication and more gateways.

V. RESULTS

A. The Unified Flexible Model

The resulting model is shown in Fig. 3. It contains one start event, one end event, nine tasks and eight XOR gateways arranged as four split/join pairs, connected by 22 sequence flows. The two shared tasks form the backbone on either side of the first variation point, and the variation points follow one another in the order credit source, assessment mode, decision and archiving.

B. Path Coverage

Table III shows the outcome of tracing each variant. In every case the activity sequence produced by the unified model is identical to that of the original rigid variant, so all four variants remain executable.

C. Structural Comparison

Table IV compares the four separate rigid models with the unified model. The first four rows reproduce the figures from the project's initial analysis; the last row was obtained by counting sequence flows in the BPMN files (each rigid variant has one more flow than it has activities).

Eliminating duplicated activity boxes removes 13 of the 22 activity nodes. The total node count, which also includes events and gateways, falls from 30 to 19, a smaller gain of about 37%, because the gateways partly offset the savings. The gain in sequence flows is smaller still (about 15%), which indicates that most of the benefit lies in maintaining each shared activity once rather than in drawing fewer connections.

TABLE IV
SEPARATE RIGID MODELS VS. UNIFIED FLEXIBLE MODEL

Metric	4 rigid	1 unified	Change
Process diagrams	4	1	-75%
Activity nodes	22 (6+5+6+5)	9	-59%
Start and end events	8	2	-75%
XOR gateways	0	8	$+8$
Total nodes	30	19	-37%
Sequence flows	26	22	-15%

D. Summary of Findings

- All four variants are preserved by the unified model.
- Shared steps are defined and maintained in a single place.
- Alternatives and the optional step are distinguishable by their position and labels: both use XOR gateways, but only the optional pattern has a direct bypass flow.
- The reduction in duplication is paid for by eight gateways and four guard conditions.

VI. DISCUSSION AND LIMITATIONS

Gateway density. Eight XOR gateways make the unified diagram visually busier than any single rigid model, even though it is smaller than their union.

Over-permissive structure. Four independent binary choices allow $2^4 = 16$ structural routes, whereas only four are real business variants. The guard conditions on the `variant` variable restrict execution to the intended four, but the diagram alone does not express this restriction. Explicit guidance, such as the allowed/discouraged/denied classification of CF4BPMN [3], would make the intent visible to the reader.

Scope. Only one case, a loan application process, is studied, and the comparison counts model elements rather than measuring comprehension or maintenance effort. The results therefore show reduced duplication, not improved usability.

Out of scope. Flexibility by adaptation, i.e., changing models or running instances at run time [1], is not addressed.

VII. CONCLUSION AND FUTURE WORK

This report showed that optional and alternative activities, expressed through XOR gateways with labelled and guarded branches, are sufficient to merge four loan-application variants into a single BPMN model without losing any variant. The unified model reduces the number of diagrams from four to one and the number of activities from 22 to 9, while introducing eight gateways. The evidence is structural and limited to a single illustrative case.

Two extensions are planned: (i) annotating the unified model with CF4BPMN-style guidance to document which choices are allowed, discouraged or denied; and (ii) applying the approach to a second domain, university admissions, to test whether the patterns generalise. A user study on readability and maintenance effort would further complete the evaluation.

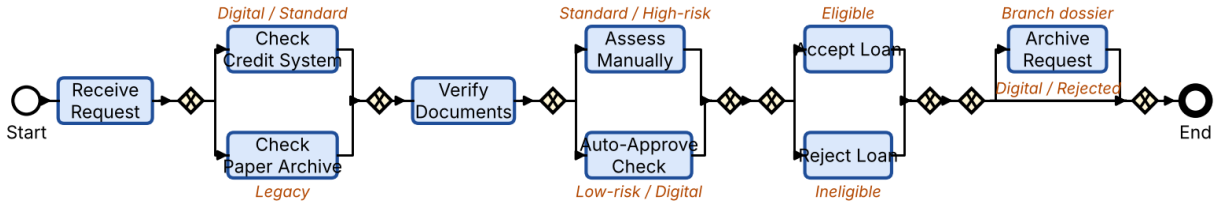


Fig. 3. Unified flexible BPMN model covering variants V1–V4. Each split/join pair of diamonds is a variation point; edge labels give the business meaning of each branch.

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